Is It the Model or the Reading?
A Fast-Weight View of ViT Attention Maps

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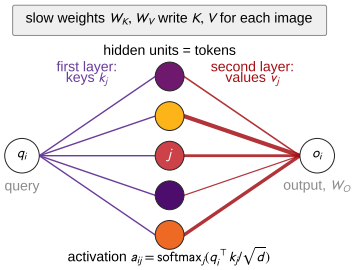
Abstract

Deep-layer attention maps of vision transformers (ViTs) often look poor: scattered, with much score on a few background patches. The usual remedy changes the model; we ask how much of the problem lies in how the map is read. A long-standing fast-weight view treats an attention head as a two-layer network with the tokens as hidden units. We state three premises of reading these units, measure a diagnostic for each on DeiT-S, DeiT-B and DINO ViT-S, and test a correction against a matched control. Our pre-registered prediction that the three corrections together improve deep-layer maps holds in one of four cases. Singly, contribution weighting, in practice the value-norm weighting of Kobayashi et al. and Zhu et al., improves localisation and human alignment in the two models its diagnostic flags (object enrichment $+0.038$ and $+0.065$) and nothing in the third, while relocating scores with a position probe harms every map, so this diagnostic may not measure its premise. On DeiT-S, two model-side changes do not exceed our margin. Part of what makes deep maps look poor is how we read them.

1 Introduction

Attention maps are widely used to show what a vision transformer (ViT) [13] looks at [1, 8, 10]. In deep layers they often look poor: the score is scattered, much of it on a few background patches [11] (Fig. 1b). The usual remedy changes the model: register tokens absorb the stray attention [11, 17]. We ask instead: *how much of the problem lies in how the map is read?*

(a) The lens



(b) Example maps (DeiT-S, layers 9–12)

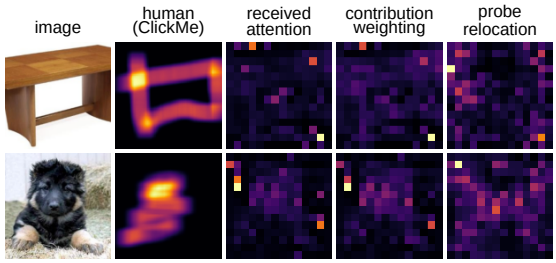


Figure 1: (a) For one image, an attention head is a two-layer network: slow weights W_K, W_V write its weights (keys, values); tokens are hidden units. (b) Two randomly drawn ClickMe images (min–max normalised).

A map is a model plus a reading rule, and the rule is rarely examined. We view an attention head as a two-layer network that is written anew for each image and has the tokens as its hidden units (Fig. 1a), a long-standing fast-weight view. The common *received-attention* map then reads these hidden units, which rests on three premises: activations are specific to the image, unit j still stands for patch j, and a large activation is a large contribution. On DeiT-S, DeiT-B [25] and DINO ViT-S [8], we measure a

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diagnostic for each premise per layer, test a correction against a matched control, and compare the outcomes with pre-registered predictions. Our contributions are:

(1) **Premises.** Three testable premises of the attention-map reading, each with a diagnostic.
(2) **A reading-side repair.** Value-norm weighting [18, 31] improves localisation and human alignment in the two models that its diagnostic flags, and nothing in the third.
(3) **Negative results.** The registered prediction that the corrections add up holds in one of four cells, and probe relocation harms every map: the probe does not measure the location premise itself.

2 Related work

2.1 How attention maps are read and scored. Whether attention explains predictions is debated [16, 28]. Later readings add layers [1], relevance propagation [9], gradients [10], vector norms [18, 31] or statistical filtering [3]. Scoring has traps: plausible maps can ignore the model [2], deletion depends on the occlusion strategy [5], and random CNNs give centred saliency [19] (largely absent in ViTs). Most compare readings by *results*; Kobayashi et al. [18] question our third premise.

2.2 Why deep ViT maps degrade. Deep maps put much attention on high-norm background tokens, which registers absorb [11, 17]; such tokens resemble heads that learn to do nothing [6], and supervision shapes attention [26]. Tokens also mix: in language models their identity is largely kept but fades [7], and in CLS-trained ViTs last-layer tokens still resemble their own input patch [21].

2.3 Attention as a fast-weight network. Fast weights are weights that one network writes for another [24]; linear attention is such a program [23], and softmax attention is the update rule of a modern Hopfield memory [22]. This view is not ours and has mostly guided model design, as in MiTA [27]; to our knowledge, it has not been used to test how attention maps are read.

3 Method

3.1 The lens. In layer ℓ, the normalised tokens $x_1, \ldots, x_{N+1}$ (CLS and $N{=}196$ patches) enter H heads. Head h forms $q_i^h{=}W_Q^h x_i$, $k_j^h{=}W_K^h x_j$ and $v_j^h{=}W_V^h x_j$, and the block adds to token i

$$o_i = \sum_h W_O^h \sum_j a_{ij}^h \, v_j^h, \qquad a_{ij}^h = \mathrm{softmax}_j\big(q_i^{h\top} k_j^h / \sqrt{d}\big). \tag{1}$$

For a fixed image this is a two-layer network applied to q_i: token j is a hidden unit with input weights k_j^h, activation a_{ij}^h and output weights $W_O^h v_j^h$. As the softmax makes units compete, the network is a guiding approximation, not an exact equivalence. *Received attention* sums each unit's activation over queries and heads, $s_j = \sum_h \sum_i a_{ij}^h$, and draws it at patch j. As our question concerns deep layers, maps sum s_j over layers 9–12 (the last third), a range set before the experiments.

3.2 From reading units to premises. If a head is a network, an attention map *reads its hidden units*, and three lessons about reading units each give a premise (Table 1): what is there for every input says nothing about this input; what a unit stands for must be measured, not read off its index [4]; and a large activation is not a large effect, because output weights count too [30]. No single premise needs the lens (the third is the point of Kobayashi et al. [18]); it gives the list and adds the location premise, which the readings of Section 2 all assume. Each premise gets a diagnostic, a correction and a matched control that changes the map alike without the premise's information. Diagnostics are proxies, not the premises themselves, and c_j measures the size of a token's output, not its effect on the prediction: norms ignore cancellation between heads and all later layers. A correction counts only if it beats its control; it should help where its diagnostic flags a failure and change little elsewhere.

3.3 Metrics. A centre prior can inflate scores: in our data, a centred Gaussian has a Spearman correlation of 0.41 with human ClickMe maps. Two metrics are therefore centre-blind: *same-ring object enrichment*, the object rate (ImageNet-S masks [15]) of the top-scoring patches minus that of all patches equally far from the centre, and *image-specific ClickMe alignment* [14, 20], the Spearman correlation with the image's own human map minus that with another image's. *Deletion faithfulness* [5] is not centre-blind. A paired difference is *higher* or *lower* only if its 95% bootstrap interval excludes zero and it exceeds a margin (0.02; 0.05 for deletion); else it is *comparable*.

Table 1: The premises, their tests, and the outcome in layers 9–12 on the centre-blind metrics (S, B, D: DeiT-S, DeiT-B, DINO ViT-S). *One gain, equal to its control's. †On B ClickMe, not beyond its control.

Premise	Diagnostic (a proxy)	Correction	Matched control	Flagged	Helps
Specificity: activations are image-specific	variance explained by the mean map $\bar{s}$	mean-map subtraction, $s - \bar{s}$	centred Gaussian subtraction	none	none*
Location: unit j still stands for patch j	probe accuracy for the patch index of x_j	probe relocation, $r_p = \sum_j s_j P(p \mid x_j)$	blur of equal displacement	S, B, D	none
Contribution: large activation, large effect	share of s on high-norm tokens	contribution weighting, $c_j = \sum_{h,i} a_{ij}^h \lVert W_O^h v_j^h \rVert$	zero the high-norm tokens	S, B	S, B†

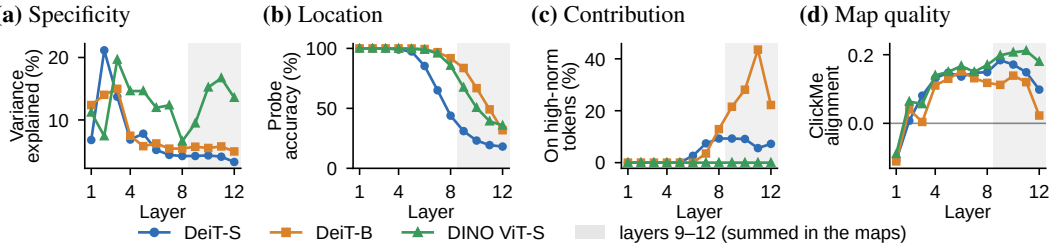


Figure 2: Per layer: (a) map variance explained by the mean map; (b) position-probe accuracy; (c) attention on high-norm tokens; (d) image-specific ClickMe alignment of single-layer maps. Bands: 95% intervals.

4 Experiments

4.1 Setup and registered predictions. We use frozen DeiT-S and DeiT-B [25] (supervised) and DINO ViT-S/16 [8]; the model-side changes are a test-time register [17] and the top-k-attention checkpoint of MiTA [27], both for DeiT-S. Data: 5,000 ImageNet [12] and 12,419 ImageNet-S images, 17,408 ClickMe maps and 4,000 held-out images (App. A). Before the experiments we fixed metrics, margins and comparisons and registered three predictions (App. A). **R1:** where a diagnostic shows no failure, the correction leaves the map comparable (contribution weighting in DINO ViT-S; relocation at layers 1–4). **R2:** the three corrections together are higher than received attention in the four supervised centre-blind cells. **R3:** neither model-side change is higher than DeiT-S. For single corrections against their controls we fixed no direction. Other analyses are *post hoc*.

4.2 Attention marks the units the model uses. If, in every head and layer, each query reads only the $k{=}16$ units (of 196) that score highest for its region, the models keep 72%, 83% and 61% of their top-1 predictions, against 9%, 30% and 4% for random units, also for other k (App. B).

4.3 Two diagnostics flag deep layers (Fig. 2). *Specificity* shows no failure: in layers 6–12 the mean map explains 3–6% of a map's spatial variance in the supervised models and at most 17% in DINO ViT-S, and in layers 9–12 it is 82–98% radial. The *location* diagnostic weakens in all models: probe accuracy falls from nearly 100% in layers 1–5 to 18%, 32% and 36% at layer 12 (chance: 0.5%), although 45%, 66% and 81% of its probability lies in the $3{\times}3$ neighbourhood of the true patch. The *contribution* diagnostic flags only the supervised models: high-norm tokens (norm over three times the image median) get up to 44% of the attention in DeiT-B and 9% in DeiT-S but none in DINO ViT-S [11], with value vectors 0.29 and 0.23 times as long as other patches'.

4.4 Only contribution weighting improves the map (Table 2). *R1 holds:* in DINO ViT-S, contribution weighting leaves all metrics comparable, and relocation at layers 1–4 changes the centre-blind metrics by at most 0.004. *R2 holds in one of four cells* (DeiT-B ClickMe). *Contribution weighting* is higher on both centre-blind metrics in DeiT-S and DeiT-B ($+0.033$ to $+0.065$), beating its control except on DeiT-B ClickMe, and comparable on deletion: localisation and human alignment improve, deletion faithfulness stays within the margin. In DeiT-B, zeroing the high-norm tokens gives $+0.040$ of the $+0.065$ on enrichment. Post hoc, it beats the same weights shuffled within each head in all four cells ($+0.035$ to $+0.073$), gains more on images with more attention on high-norm tokens, and is matched by weighting with $\lVert v_j \rVert$ alone (App. B): in practice it is value-norm weighting [18, 31]. At layer 12 it recovers 31–62% of the drop in map quality (Fig. 2d). *Mean-map subtraction* gains

Table 2: Layers 9–12, controls indented: enrichment (Enr.), ClickMe (Click.), deletion (Del.). $^{\uparrow}/^{\downarrow}$: higher/lower than received attention (model-side: than DeiT-S). GAE [10] also uses gradients; it has no significance test.

	DeiT-S			DeiT-B			DINO ViT-S		
Reading	Enr.	Click.	Del.	Enr.	Click.	Del.	Enr.	Click.	Del.
Received attention	0.140	0.157	0.236	0.080	0.107	0.169	0.199	0.208	0.285
Contribution weighting	$0.178^{\uparrow}$	$0.190^{\uparrow}$	0.253	$0.146^{\uparrow}$	$0.165^{\uparrow}$	0.198	0.196	0.207	0.285
zero high-norm tokens	0.152	0.168	0.241	$0.121^{\uparrow}$	$0.148^{\uparrow}$	0.186	0.199	0.208	0.285
Mean-map subtraction	0.138	0.165	0.208	0.077	$0.087^{\downarrow}$	0.144	0.201	$0.252^{\uparrow}$	$0.233^{\downarrow}$
centre-Gaussian subtraction	0.133	0.168	$0.167^{\downarrow}$	$0.056^{\downarrow}$	$0.062^{\downarrow}$	$0.024^{\downarrow}$	0.200	$0.253^{\uparrow}$	$0.232^{\downarrow}$
Probe relocation	$0.041^{\downarrow}$	$0.084^{\downarrow}$	$0.076^{\downarrow}$	$0.027^{\downarrow}$	$0.050^{\downarrow}$	$0.051^{\downarrow}$	$0.084^{\downarrow}$	$0.143^{\downarrow}$	$0.141^{\downarrow}$
matched blur	$0.095^{\downarrow}$	0.152	$0.181^{\downarrow}$	$-0.008^{\downarrow}$	$0.068^{\downarrow}$	$0.097^{\downarrow}$	0.189	0.226	0.287
All three corrections	$0.059^{\downarrow}$	$0.123^{\downarrow}$	$0.076^{\downarrow}$	0.079	$0.137^{\uparrow}$	$0.085^{\downarrow}$	$0.088^{\downarrow}$	$0.177^{\downarrow}$	$0.090^{\downarrow}$
+ test-time register	0.152	0.169	0.236	–	–	–	–	–	–
Top-k training (MiTA)	$0.100^{\downarrow}$	$0.115^{\downarrow}$	–	–	–	–	–	–	–
GAE (reference)	0.204	0.234	0.345	0.196	0.213	0.308	0.221	0.234	0.418

only on DINO ViT-S ClickMe, as much as its control. *Probe relocation* is lower in all nine cells, and in DeiT-S and DINO ViT-S lower even than its matched blur. *R3 holds:* the register gains $+0.012$ and $+0.013$, below the margin; the top-k checkpoint is lower, but differs from DeiT-S in training and preprocessing, so the comparison is uncontrolled.

4.5 Does the diagnosis tell where a correction helps (Table 1, right)? We registered no direction for single corrections, so this reading is post hoc. For *contribution* it is consistent: the correction helps in the two flagged models and changes nothing in DINO ViT-S. For *specificity* there was little to correct, and nothing is gained beyond the control. For *location* it is not: the diagnostic weakens in every model, yet relocation harms every map. Four explanations, such as far probe errors or a weak probe, found no support (post hoc; App. C). A fifth, untested, concerns the diagnostic: the probe measures whether a token's patch index can be decoded, not where its content comes from. A token may lose its position code and still carry patch j (cf. [21]); the index then stays the better guide. Diagnosis and correction agree for one premise in three models, which does not establish the idea.

5 Discussion

5.1 Interpretation. *Activation is not effect:* high-norm tokens draw attention but have short value vectors, like heads that do nothing [6]; weighting by output-weight norms partly repairs the map without changing the model. *A weaker position code is not a moved token:* to test the location premise itself, one must measure which input patches a token depends on.

5.2 Limitations. We study three classifiers at S/B scale, each trained once, and only received attention; CLS attention, larger or register-trained models and model-side changes beyond DeiT-S are untested. Diagnostics are proxies, and the high-norm threshold was not varied. ClickMe images were very likely training data. Gains are modest, hard to see in one image (Fig. 1b), and below GAE's.

5.3 Ethical considerations. We use public data and checkpoints; the images include people and known biases. An attention map, corrected or not, is only partial evidence about a model.

6 Conclusion

For ViT attention maps, the fast-weight view gives three premises to check: correcting the contribution premise partly repairs deep maps in the models that its diagnostic flags, the location correction fails, and the corrections do not add up. We tested only ViT attention maps, but the same question applies to any method that maps internal values to input positions. Future work should test the location premise with token-to-input attribution and cover larger and register-trained models and CLS attention. As a broader impact, premise tests need no retraining and can keep users from over-trusting a map.

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A Experimental details

Models. All models are frozen ViTs with 16×16 patches on 224×224 inputs, taken from timm [29]. *DeiT-S and DeiT-B* [25]: `deit_small_patch16_224.fb_in1k` and `deit_base_patch16_224.fb_in1k` (resize to 248, centre crop 224: a crop ratio of 0.9). *DINO ViT-S/16* [8]: `vit_small_patch16_224.dino` with the official DINO linear head (resize to 256, centre crop 224: a crop ratio of 0.875). *DeiT-S with a test-time register* [17]: the activations of eight feed-forward units in layers 4 and 5 are moved to one added register token, which is left out of every map, like CLS. The units were selected on 500 ImageNet-S images, where they reduce the number of high-norm patch tokens (counted per layer) by 92.75% and keep 99.0% of the predictions; on the other 11,919 images the reduction is 91.5%, and ImageNet accuracy stays at 79.8%. For DeiT-B the same search found no such units (44.9% reduction with 32 units), so we report the register for DeiT-S only. *Top-k training*: the DeiT-S checkpoint released with MiTA [27] (25 summary slots, $k = 25$), with its own preprocessing (crop ratio 0.875). It differs from our DeiT-S in training and preprocessing, so its row in Table 2 is not a controlled comparison; its deletion score is undefined because its summary slots pool fixed regions of the full grid. Both model-side fixes are tested at size S only; in a preliminary comparison with an earlier protocol, top-k training scored 0.011 lower in object enrichment than its baseline at size S but 0.037 higher at size Tiny, so the result should not be read beyond DeiT-S.

Data. *Execution test*: 5,000 randomly drawn ImageNet validation images [12]. *ImageNet-S* [15]: all 12,419 validation images with segmentation masks; enrichment is defined for 12,150 of them (12,144 for DINO ViT-S), and deletion uses a random subset of 2,000. *ClickMe* [14, 20]: 17,408 images with human importance maps, which we smooth with a Gaussian. *Held-out set*: 4,000 ImageNet validation images that are not evaluated anywhere; 2,000 train the position probes, and the other 2,000 give the mean maps, the probe accuracies and the blur widths. While building this set we found that only 2 of 4,002 candidate validation images matched a ClickMe image, so the ClickMe images are very likely ImageNet training images, which all three models have seen.

Diagnostics. All three are computed per layer. *Specificity*: with $\bar{s}$ the mean received-attention map of the held-out images and Var_j the variance over the 196 patches of one image, the share explained is $1 - \sum_x \mathrm{Var}_j(s - \bar{s}) \,/\, \sum_x \mathrm{Var}_j(s)$, summed over the ImageNet-S images x. *Location*: the top-1 accuracy of the linear position probe on the 196 patch tokens of each of 2,000 held-out images. *Contribution*: the share of received attention on high-norm tokens, on the ImageNet-S images. The high-norm threshold (three times the median, defined below) is an operational definition fixed before the experiments; we did not test other thresholds. The correction itself uses no threshold.

Corrections and matched controls. All are applied per layer and then summed over layers 9–12; queries run over all 197 tokens and maps cover the 196 patches. *Mean-map subtraction* subtracts the model's mean received-attention map over the 2,000 held-out images. Its control subtracts a centred isotropic Gaussian ($\sigma = 3.5$ patches) scaled to the spatial standard deviation of that mean map; it is exactly radial. *Probe relocation* uses one linear 196-way classifier per model and layer (36 in all) on the standardised tokens after the block's first layer norm (Adam, learning rate 0.01, 30 epochs) and computes $r_p = \sum_j s_j P_\ell(p \mid x_j)$, which keeps the total score. Its control is a Gaussian blur whose width is set so that it moves the score as far, in mean squared displacement, as relocation does on held-out tokens of that layer (0.7–2.9 patches). *Contribution weighting* computes $c_j = \sum_h \sum_i a_{ij}^h \|W_O^h v_j^h\|_2$, where W_O^h is the head's slice of the output projection; $a_{ij}^h \|W_O^h v_j^h\|$ is the per-head norm of Kobayashi et al. [18]. A *high-norm token* is a patch whose block input has a norm above three times the image's median patch norm in that layer; the control sets received attention to zero at these patches. *All three corrections*: contribution weighting, then probe relocation, then subtraction of the held-out mean of the resulting map.

Metrics. *Same-ring object enrichment.* A patch is an object patch if at least half of its pixels belong to the image's class mask. We select as many top-scoring patches as there are object patches. For every ring of patches at the same distance from the grid centre, we take the object rate among its selected patches minus that among all its patches, and average over rings, weighted by their selected patches. A purely radial map scores zero in expectation. *Image-specific ClickMe alignment.* The map is upsampled to 224×224 and correlated (Spearman, over pixels) with the image's own human map; we subtract its correlation with the human map of another image. A map that does not depend

on the image scores exactly zero. *Deletion faithfulness.* Patches are removed in the order of the map by dropping their tokens; at budgets of 0 to 147 patches we record whether the top-1 class is unchanged. The score is the symmetric relevance gain of Blücher et al. [5]: the area under the least-important-first curve minus that under the most-important-first curve. *Validity of the metrics.* A random map, a centred Gaussian and the mean map all score within 0.004 of zero on enrichment and on ClickMe alignment (the two image-independent maps score exactly zero on the latter); on deletion the centred Gaussian scores 0.110, 0.099 and 0.140 (DeiT-S, DeiT-B, DINO ViT-S), which is why every correction is compared with a matched control. *Statistics.* One value per image; paired bootstrap over images (1,000 resamples, 95% percentile intervals). A difference is *higher* (*h*) or *lower* (*l*) if its interval excludes zero and its size exceeds 0.02 (0.05 for deletion), and *comparable* (*c*) otherwise. Intervals are not adjusted for multiple comparisons; a verdict needs the margin as well.

Registered predictions. Before the formal runs we fixed the definitions above, the margins, the comparisons of Table 2 and the predictions of Table A.1; the record is the file `docs/preregistration.md` of the repository. R2 was registered with three possible verdicts: holds (four of four cells), partly holds (one to three) and does not hold (none). For each correction against its control, the comparison was fixed without a direction. Earlier runs of the execution test and of the diagnostics had been seen, so the last two rows are replications. Other analyses are marked post hoc.

Table A.1: Registered predictions and outcomes (centre-blind metrics: enrichment and ClickMe alignment).

	Prediction	Outcome
R1	Contribution weighting is comparable to received attention in DINO ViT-S on all three metrics	holds (3 of 3 cells)
R1	Relocation at layers 1–4 is comparable to received attention on the centre-blind metrics in all models	holds (6 of 6 cells)
R2	All three corrections together are higher than received attention on the centre-blind metrics in DeiT-S and DeiT-B	partly holds (1 of 4 cells: DeiT-B, ClickMe)
R3	Neither model-side change is higher than DeiT-S received attention on the centre-blind metrics	holds (4 of 4 cells)
–	Execution agreement rises with k and exceeds that of random sets at every k	holds in all models
–	Probe accuracy falls with depth; the share of attention on high-norm tokens peaks in layers 6–12 of the supervised models and is below 1% in every layer of DINO ViT-S	holds in all models

Compute and code. All experiments took about 19 GPU hours on NVIDIA L4 GPUs. Code and result tables: `https://github.com/hhuang999/vit-fastweight-lens`.

B Additional results

Execution test. The 14×14 grid is pooled into 16 overlapping regions. In every layer and head, each region's mean query is scored against all keys, and its k highest-scoring keys form the region's reading set (the CLS query keeps its own top-k set). The sets come from one pass of the unmodified model. We then rerun the model with every query restricted to its region's set, in all 12 layers at once, and record whether the top-1 class is unchanged. Figure B.1 shows that top-k sets beat random sets of the same size at every k from 4 to 128 in all three models, and that agreement rises with k. The dotted line alone was run with a crop ratio of 0.875 for all models (0.9 for DeiT-S and DeiT-B elsewhere).

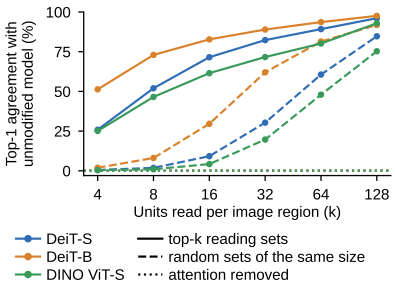


Figure B.1: Top-1 agreement with the unmodified model when queries read only top-k units (solid), random units (dashed) or nothing (dotted). 95% intervals: at most ±1.5 points.

Table B.1: Diagnostics per layer, as plotted in Fig. 2 (S, B, D: DeiT-S, DeiT-B, DINO ViT-S; probe chance: 0.5%).

	Specificity: mean-map share (%)			Location: probe accuracy (%)			Contribution: high-norm share (%)			Map quality: ClickMe alignment		
Layer	S	B	D	S	B	D	S	B	D	S	B	D
1	6.8	12.4	11.2	100.0	100.0	99.9	0.0	0.0	0.0	−0.108	−0.111	−0.089
2	21.2	14.0	7.4	100.0	100.0	100.0	0.0	0.0	0.0	0.007	0.041	0.064
3	13.7	15.0	19.7	99.9	100.0	99.9	0.0	0.0	0.0	0.081	0.004	0.056
4	6.9	7.5	14.6	99.0	100.0	99.8	0.0	0.0	0.0	0.132	0.110	0.140
5	7.8	5.8	14.6	97.5	99.9	99.6	0.1	0.0	0.0	0.148	0.130	0.150
6	5.2	6.2	11.9	85.3	99.5	98.9	2.6	0.0	0.0	0.136	0.151	0.167
7	4.4	5.4	12.4	65.0	96.8	95.9	7.4	3.5	0.0	0.147	0.131	0.150
8	4.2	5.4	6.6	43.9	91.8	85.6	9.3	12.9	0.0	0.149	0.118	0.170
9	4.2	5.7	9.5	31.0	83.6	67.4	9.3	21.5	0.0	0.184	0.112	0.198
10	4.3	5.5	15.2	23.1	66.8	50.5	9.1	28.1	0.0	0.171	0.139	0.207
11	4.1	5.8	16.7	19.4	49.2	39.2	5.6	43.5	0.0	0.149	0.120	0.211
12	3.3	5.0	13.6	18.1	31.8	35.7	7.2	22.2	0.0	0.098	0.023	0.179

Diagnostics per layer. Table B.1 lists the values plotted in Fig. 2. Two numbers of the main text are not in the table. First, in layers 9–12, 82–98% of the variance of the mean map is explained by the distance from the grid centre. Second, averaged over layers 9–12, the mean value-vector norm $\|v_j\|$ of high-norm tokens is 0.29 (DeiT-S) and 0.23 (DeiT-B) times the median of the other patches (measured on 11,919 ImageNet-S images).

Recovery at the last layer. In the supervised models, single-layer received attention is worse at layer 12 than at the best of layers 8–11 (Fig. 2d). Applied to layer 12, contribution weighting recovers 62% and 33% of this drop on enrichment (DeiT-S, DeiT-B), 31% and 45% on ClickMe alignment, and 8% and 29% on deletion. The other two corrections make layer 12 worse.

Differences behind Table 2. Table B.2 gives the paired differences with their intervals. The values in Table 2 have 95% intervals of at most ± 0.005 on enrichment and ClickMe alignment and ± 0.012 on deletion. The model-side differences from DeiT-S received attention are $+0.012$ [0.011, 0.013] on enrichment, $+0.013$ [0.012, 0.013] on ClickMe and 0.000 [-0.005, 0.004] on deletion for the test-time register, and -0.040 [-0.042, -0.039] and -0.042 [-0.044, -0.040] on the first two for top-k training.

Tests of the contribution account. Contribution weighting should work by down-weighting tokens that receive attention but contribute little. One consequence was registered (R1), and two further tests were added post hoc. *No gain without high-norm tokens (R1).* In DINO ViT-S, where no attention falls on high-norm tokens, every difference from received attention is at most 0.004 in size (Table B.2). *More high-norm attention, more gain (post hoc).* Across the images of a model, the gain grows with the share of received attention on high-norm tokens: Spearman $+0.075$ [0.057, 0.092] and $+0.132$ [0.113, 0.150] on enrichment (DeiT-S, DeiT-B), $+0.045$ [0.032, 0.060] and $+0.158$ [0.143, 0.173] on ClickMe. *The weights must match the tokens (post hoc).* Against the same weights shuffled among the patches of each head, contribution weighting is higher in all four supervised cells: $+0.038$ [0.037, 0.040] and $+0.073$ [0.071, 0.076] on enrichment, $+0.035$ [0.035, 0.036] and $+0.067$ [0.065, 0.068] on ClickMe. Finally, also post hoc, weighting by $\|v_j^h\|$ instead of $\|W_O^h v_j^h\|$ is comparable in every cell (differences -0.010 to 0.000): in practice the correction is the value-norm weighting of Kobayashi et al. [18] and Zhu et al. [31].

C Why does probe relocation fail?

Relocation lowers every deep-layer cell (Table 2). Two planned comparisons narrow the problem. Where the probe is accurate, relocation is harmless (R1): at layers 1–4 it changes enrichment and ClickMe alignment by at most 0.004 in every model. And in DeiT-S and DINO ViT-S it is lower than a blur that moves the score equally far (Table B.2). The rest of this section is post hoc: Table C.1 lists four explanations that we tested after seeing this result; none is supported. In particular, the probe is

Table B.2: Paired differences, layers 9–12: mean with 95% paired-bootstrap interval and verdict (*h* higher, *c* comparable, *l* lower). Top: against received attention. Bottom: each correction against its matched control.

Model	Same-ring enrichment	ClickMe alignment	Deletion faithfulness
Contribution weighting − received attention			
DeiT-S	0.038 [0.036, 0.039] *h*	0.033 [0.032, 0.034] *h*	0.017 [0.013, 0.021] *c*
DeiT-B	0.065 [0.063, 0.067] *h*	0.058 [0.056, 0.059] *h*	0.029 [0.024, 0.034] *c*
DINO ViT-S	−0.004 [−0.004, −0.003] *c*	−0.001 [−0.001, −0.001] *c*	0.000 [−0.003, 0.004] *c*
Mean-map subtraction − received attention			
DeiT-S	−0.002 [−0.003, −0.002] *c*	0.009 [0.008, 0.010] *c*	−0.028 [−0.034, −0.023] *c*
DeiT-B	−0.003 [−0.004, −0.003] *c*	−0.020 [−0.021, −0.019] *l*	−0.025 [−0.031, −0.019] *c*
DINO ViT-S	0.002 [0.001, 0.003] *c*	0.044 [0.040, 0.048] *h*	−0.052 [−0.059, −0.044] *l*
Probe relocation − received attention			
DeiT-S	−0.099 [−0.101, −0.097] *l*	−0.073 [−0.076, −0.069] *l*	−0.161 [−0.171, −0.150] *l*
DeiT-B	−0.054 [−0.055, −0.052] *l*	−0.057 [−0.059, −0.055] *l*	−0.118 [−0.127, −0.110] *l*
DINO ViT-S	−0.115 [−0.117, −0.113] *l*	−0.064 [−0.067, −0.062] *l*	−0.144 [−0.153, −0.134] *l*
All three corrections − received attention			
DeiT-S	−0.081 [−0.083, −0.079] *l*	−0.034 [−0.038, −0.030] *l*	−0.160 [−0.171, −0.149] *l*
DeiT-B	−0.001 [−0.003, 0.001] *c*	0.030 [0.027, 0.032] *h*	−0.084 [−0.093, −0.076] *l*
DINO ViT-S	−0.111 [−0.114, −0.109] *l*	−0.031 [−0.035, −0.027] *l*	−0.195 [−0.205, −0.184] *l*
Contribution weighting − zero high-norm tokens (its control)			
DeiT-S	0.026 [0.025, 0.027] *h*	0.022 [0.021, 0.023] *h*	0.012 [0.008, 0.016] *c*
DeiT-B	0.025 [0.024, 0.026] *h*	0.017 [0.016, 0.018] *c*	0.013 [0.008, 0.018] *c*
DINO ViT-S	−0.004 [−0.004, −0.003] *c*	−0.001 [−0.001, −0.001] *c*	0.000 [−0.003, 0.004] *c*
Mean-map subtraction − centre-Gaussian subtraction (its control)			
DeiT-S	0.005 [0.004, 0.005] *c*	−0.003 [−0.005, −0.001] *c*	0.040 [0.034, 0.047] *c*
DeiT-B	0.022 [0.021, 0.023] *h*	0.025 [0.020, 0.030] *h*	0.120 [0.109, 0.130] *h*
DINO ViT-S	0.001 [0.001, 0.002] *c*	−0.001 [−0.001, 0.000] *c*	0.001 [−0.003, 0.005] *c*
Probe relocation − matched blur (its control)			
DeiT-S	−0.054 [−0.056, −0.051] *l*	−0.068 [−0.073, −0.064] *l*	−0.105 [−0.116, −0.095] *l*
DeiT-B	0.035 [0.034, 0.036] *h*	−0.018 [−0.020, −0.015] *c*	−0.047 [−0.055, −0.037] *c*
DINO ViT-S	−0.105 [−0.107, −0.102] *l*	−0.083 [−0.085, −0.080] *l*	−0.146 [−0.155, −0.137] *l*

not too weak, and its errors lie mostly next to the true patch. A fifth explanation, which we could not test, is that the probe measures whether a token's position can be decoded, not where its content comes from (Section 4).

Table C.1: Four explanations for the failure of probe relocation, and their tests (post hoc). Cells are the two centre-blind metrics in the three models.

Explanation	Test (layers 9–12 unless stated)	Result
The probe sends score to distant patches	Share of the probe's error mass more than three patches from the true patch	Small: 2–29% across models and layers. At layer 12, 45%, 66% and 81% of the probability lies in the 3×3 neighbourhood of the true patch
The probe has a fixed position preference	Peak of the probe's mean prediction relative to uniform; its correlation with the mean map	Close to uniform (peak 1.1–1.3 times the uniform value); at most weakly related to the mean map
The linear probe is too weak	Two-layer MLP probe in place of the linear one	Layer-12 accuracy rises by only 1.8, 3.8 and 6.7 points; relocation is still lower than received attention in all six cells
Global tokens are harder to place	Probe accuracy and entropy against a token's similarity to the image's mean token (layers 10–12)	No consistent relation; accuracy falls with globality only in DeiT-B

D Example maps

The stored gallery holds 24 ClickMe images drawn at random with a fixed seed. Fig. 1b shows the two with the smallest ClickMe index, and Fig. D.1 the next four. Received attention and contribution weighting look much alike, while probe relocation spreads the score over many more patches. The

gains in Table 2 are averages of small changes in how each map ranks its patches, which single maps do not show.

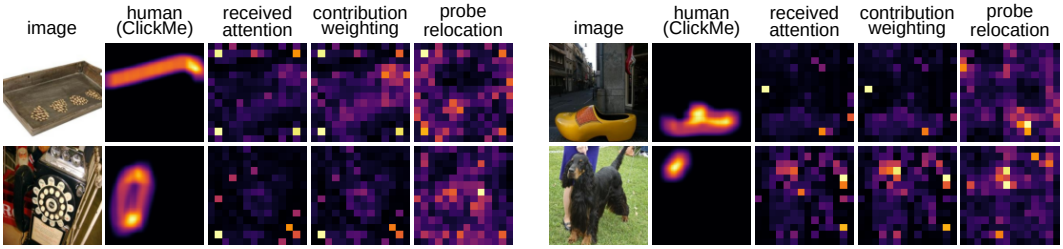


Figure D.1: DeiT-S on four further ClickMe images, layers 9–12; each map is min–max normalised.

E Key parts of the code

Given the attention a_{ij}^h and values v_j^h of a layer, the readings are:

```
def received_and_contribution(p, v, w_o, prefix=1):
    # p: [B,H,N,N] attention a_ij^h (queries i, keys j); v: [B,H,N,d] values; w_o: [D,H,d] W_O by head
    gram = torch.einsum('dhe,dhf->hef', w_o, w_o)                  # W_O^h^T W_O^h
    length = torch.einsum('bhne,hef,bhnf->bhn', v, gram, v).clamp_min(0).sqrt()
    received = p.sum(2)                                            # sum over queries i
    s = received.sum(1)[:, prefix:]                                # received attention s_j
    c = (received * length).sum(1)[:, prefix:]                     # contribution weighting c_j
    return s, c

def relocate(s, probs):                  # probe relocation r_p = sum_j s_j P(p | x_j)
    return torch.einsum('bj,bjp->bp', s, probs)
```